

Randomized and low-rank approximation

Relative-error threshold for extremal Gaussian trace bounds

Difficulty: challenging **Importance:** interesting to the community

Source: Hallman, Conjecture 3 together with Theorem 6.

Problem statement

For a real symmetric $d \times d$ matrix D and integer $m \geq 1$, define the Gaussian trace estimator

$$T_m(D) = \frac{1}{m} \sum_{j=1}^m z_j^T D z_j, \quad z_1, \dots, z_m \stackrel{\text{iid}}{\sim} N(0, I_d).$$

Let $A \neq 0$ be any real symmetric positive semidefinite $n \times n$ matrix, with $n \geq 1$, and set

$$\mu = \frac{\text{tr}(A)}{\|A\|_2}, \quad B_\mu = \frac{1}{\mu} \text{diag}(I_{\lfloor \mu \rfloor}, \mu - \lfloor \mu \rfloor).$$

Here $\|\cdot\|_2$ is the spectral norm, $\mu \geq 1$, and a zero final diagonal entry may be retained. Let X have the Gamma distribution with shape and rate both $m\mu/2$. The shape/rate convention means that $\text{Gamma}(\alpha, \beta)$ has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ for $x > 0$.

Conjecture. For every such A , every integer $m \geq 1$, and every $\varepsilon \geq 2/(m\mu)$, the complete comparison chain holds:

$$\begin{aligned} \Pr(|T_m(A) - \text{tr}(A)| \geq \varepsilon \text{tr}(A)) &\leq \Pr(|T_m(B_\mu) - 1| \geq \varepsilon) \\ &\leq \Pr(|X - 1| \geq \varepsilon). \end{aligned}$$

Each estimator uses Gaussian vectors of its own matrix dimension; only their distributions are compared.

Why it matters in numerical linear algebra

This would specify the tolerance range on which effective rank yields extremal, distribution-level confidence bounds for Gaussian trace estimation. It also applies to Frobenius-norm estimation through $\|C\|_F^2 = \text{tr}(C^T C)$.

References

1. Eric Hallman, *Extremal bounds for Gaussian trace estimation*, arXiv:2411.15454v1 (2024), §5, Theorem 6 and Conjecture 3.
2. Alice Cortinovis and Daniel Kressner, *On Randomized Trace Estimates for Indefinite Matrices with an Application to Determinants*, FoCM 22 (2022), 875–903, Theorem 1.

Status check

Theorem 6 proves the comparisons beyond an unspecified threshold; Conjecture 3 supplies the explicit threshold above. On 2026-09-08 the [arXiv record](#) still listed only v1. Title, author, conjecture-number and 2025/2026 searches found no resolution. The author's [later XTrace paper](#) studies different estimators. This is a bounded check.